1 Physical model architecture

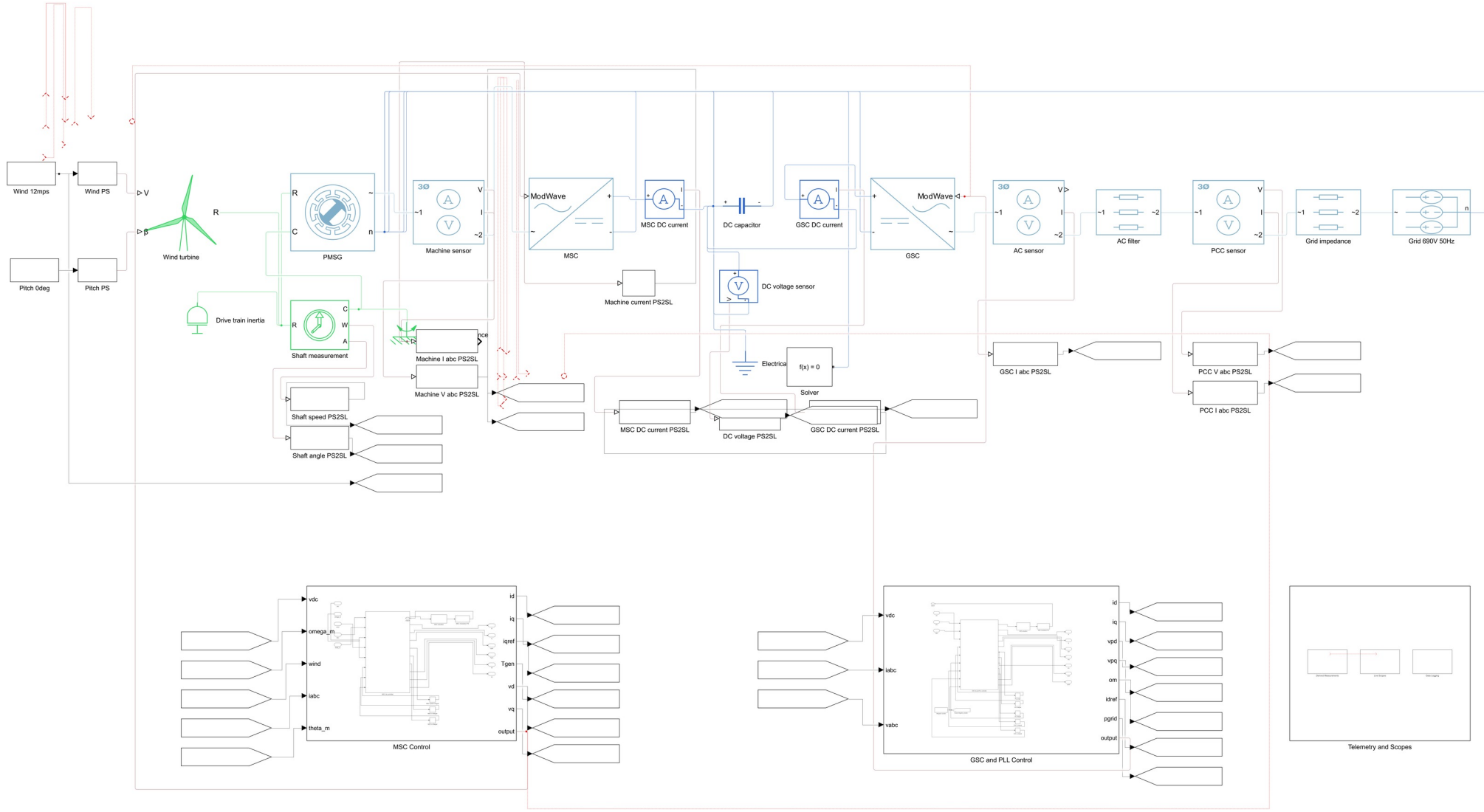


Figure 1. Full top-level model exported directly from Simulink. Power conversion runs left to right; grouped MSC and GSC/PLL controls sit below the physical network.

The saved model connects a wind turbine and shaft to a native PMSG, machine-side converter (MSC), DC capacitor, grid-side converter (GSC), RL filter, PCC and three-phase grid. The PMSM integrates stator d/q current states; both converters are averaged. The 5 MW rating follows Wang et al. (2019); other parameter values are project engineering assumptions.

2 Converter controls and weak-grid response

The MSC regulates rotor speed and machine d/q current. The GSC regulates DC-link voltage and grid current in a PLL-aligned frame. Figure 2 shows the actual control subsystems; measurement paths are separate from the physical power path.

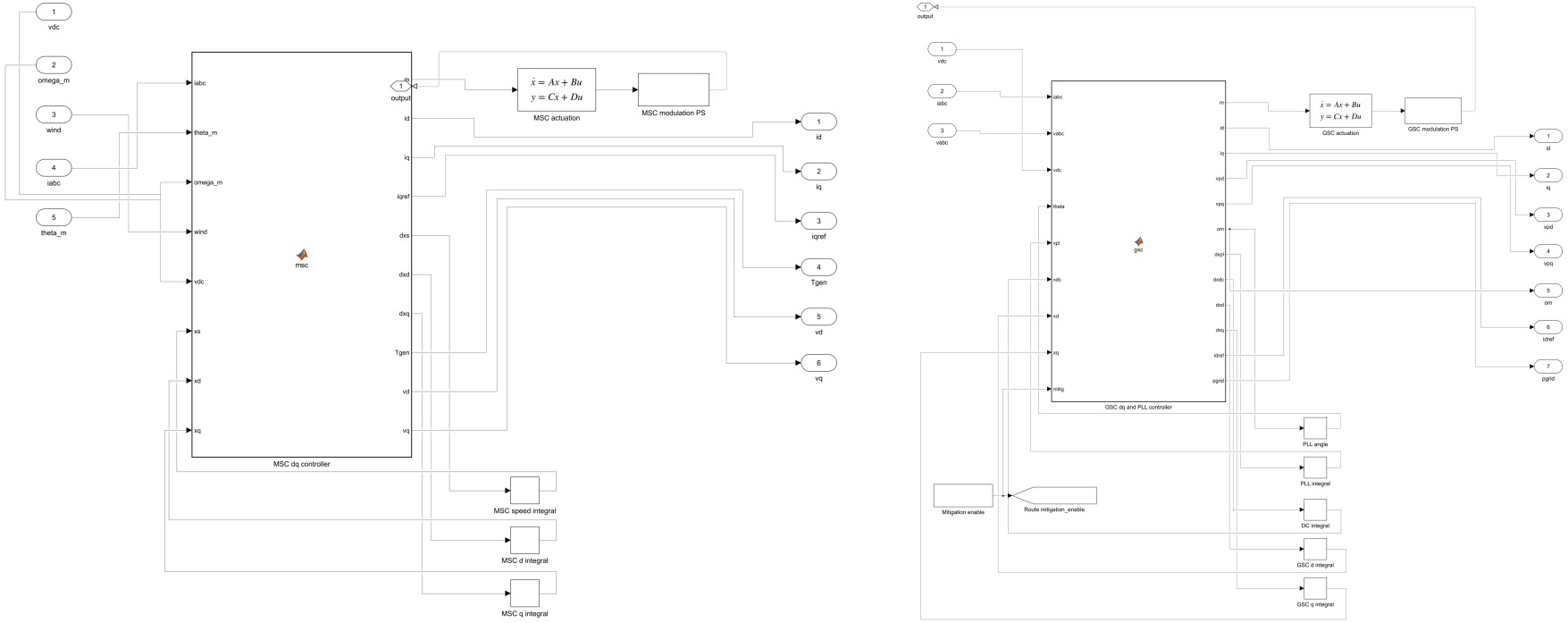


Figure 2. Machine-side control (left) and GSC/PLL control (right), exported from the saved Simulink model.

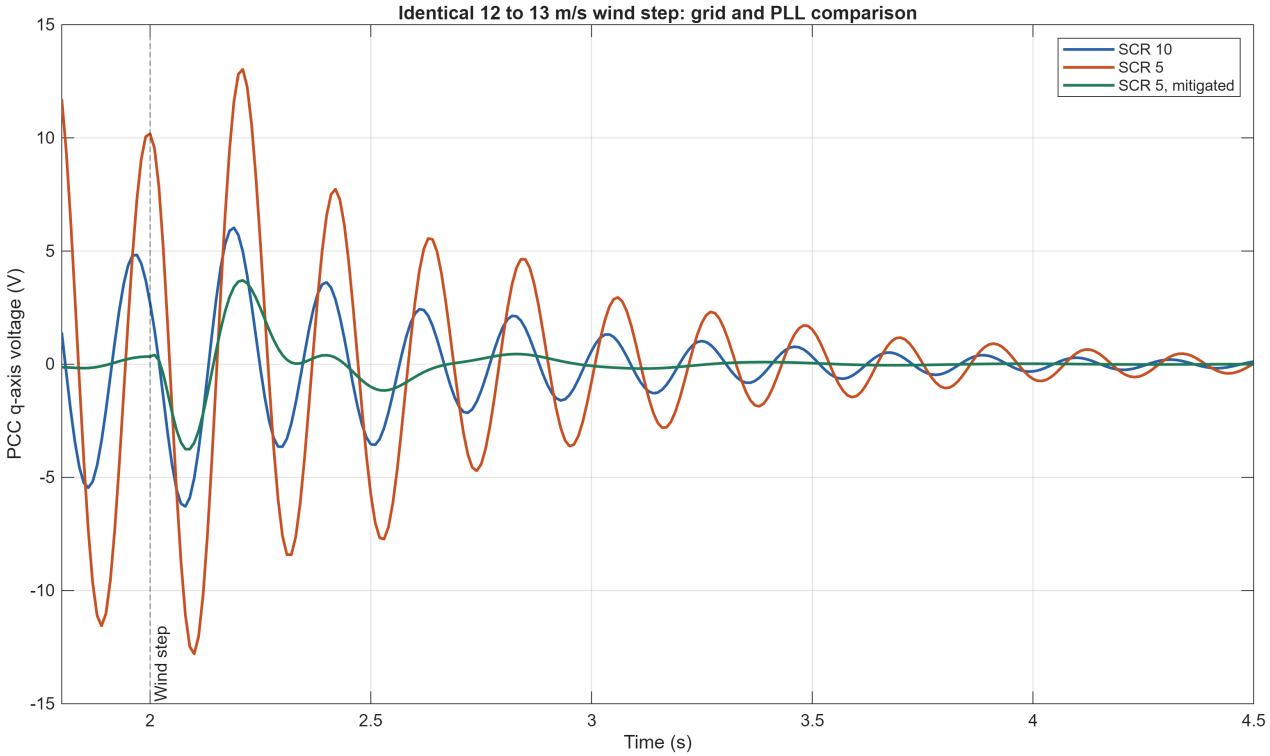
A 12 to 13 m/s wind step at 2 s changes turbine torque and converter power. Reducing nominal external short-circuit ratio (SCR) from 10 to 5 under that same step produces a larger, decaying PCC q-axis voltage oscillation. No sinusoid is injected into the model.

The weak-grid FFT peak is 4.681 Hz. The near-equilibrium physical-model snapshot mode is 4.638 Hz with 5.24% damping. The tested mitigation changes PLL proportional gain from 0.00558 to 0.02 rad/(s V) while integral gain remains 1.56 rad/(s^2 V).

Wang et al., DOI 10.1109/EI247390.2019.9061793, motivates Prony-based SSO analysis. Frequencies and gains above are measurements or assumptions of this project, not values copied from that paper.

3 Results and interpretation

Figure 3 compares identical wind steps with common voltage and time axes. Figure 4 shows an actual native Scope display after the weak-grid simulation.



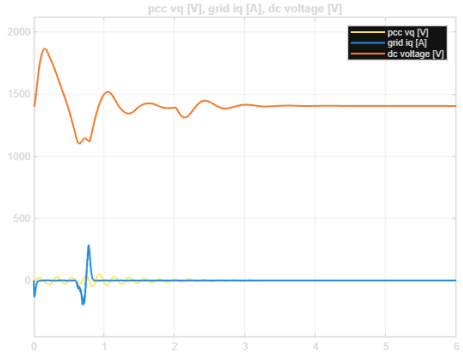


Figure 3. Measured PCC q-axis voltage comparison. Figure 4. Weak-grid Scope response from the saved physical model.

Case	Early PCC q std	Late DC voltage	Late PCC power
SCR 10 / SCR 5 / mitigated	1.788 / 3.997 / 0.554 V	1405.27 / 1405.42 / 1405.42 V	4.916 / 4.916 / 4.916 MW

The early standard deviation uses 2.25-3.25 s. PLL retuning reduces the unmitigated weak-grid metric by a factor of 7.21. Damping of the matched 4.6 Hz linearized mode rises from 5.24% to 18.99%. The mitigated least-damped mode overall is a different 1.843 Hz mode. All four scenarios and 25 physical/numerical checks pass.

The response decays. This assumed, averaged single-turbine model does not establish a sustained instability, a site-calibrated SCR threshold, PWM harmonics or shaft torsion. Full methods and parameter provenance are in the accompanying reports.